

Class exercise

How can you explain the following statement using the notion of sampling?

“On average, your friends have more friends than you do.”

What kind of assumptions (on the population and on the sample) are needed to justify this statement?

[Friendship paradox - Wikipedia](#)

The **friendship paradox** is the phenomenon first observed by the sociologist [Scott L. Feld](#) in 1991 that on average, an individual's friends have more friends than that individual.^[1] It can be explained as a form of [sampling bias](#) in which people with more friends are more likely to be in one's own friend group. In other words, one is less likely to be friends with someone who has very few friends. In contradiction to this, most people believe that they have more friends than their friends have.

Friendship Paradox

- The people with many friends constitute a smaller portion of the population.
- I am a randomly chosen person.
- I am more likely to be friends with a person who has many friends.

